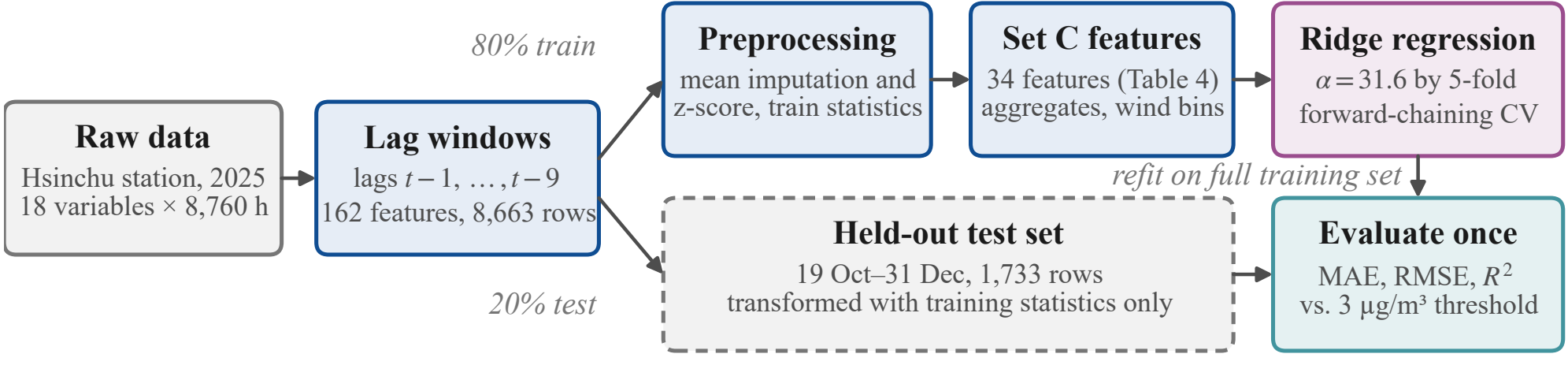

Raw data
Hsinchu station, 2025
18 variables × 8,760 h
Lag windows
lags t − 1, … , t − 9
162 features, 8,663 rows
80% train
20% test
Preprocessing
mean imputation and
z-score, train statistics
Set C features
34 features (Table 4)
aggregates, wind bins
Ridge regression
α = 31.6 by 5-fold
forward-chaining CV
refit on full training set
Held-out test set
19 Oct–31 Dec, 1,733 rows
transformed with training statistics only
Evaluate once
MAE, RMSE, R²
vs. 3 µg/m³ threshold